THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The legs of a right triangle are in the ratio $5 : 12$ and its hypotenuse is 52 cm. Its perimeter (in cm) is: 1

(a) 104 **(b)** 68 **(c)** 120 **(d)** 124

2. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 45 cm. The area of the rectangle (in cm²) is: 1

(a) 1053 **(b)** 126 **(c)** 405 **(d)** 972

3. A Pythagorean triplet whose smallest member is 15 is: 1

(a) 15, 112, 113 **(b)** 15, 113, 114 **(c)** 15, 111, 112 **(d)** 15, 16, 17

4. A rhombus has perimeter 292 cm and one diagonal 96 cm. Its area (in cm²) is: 1

(a) 5280 **(b)** 5328 **(c)** 2640 **(d)** 7008

5. A ladder 13 m long, standing in a street, rests against a house on one side when its top reaches 12 m. Without moving its foot it is turned to rest against the opposite house, reaching 5 m. The width of the street (in m) is: 1

(a) 18 **(b)** 17 **(c)** 13 **(d)** 19

6. A square has a diagonal of $6\sqrt{2}$ cm. Its area (in cm^2) is: 1

(a) 36 **(b)** 144 **(c)** 72 **(d)** 18

7. A ladder 65 m long reaches 63 m up a wall. Its top slides down 3 m along the wall. The foot of the ladder moves away from the wall by (in m): 1

(a) 16 **(b)** 3 **(c)** 9 **(d)** 25

8. The legs of a right triangle are in the ratio $8 : 15$ and its hypotenuse is 51 cm. Its perimeter (in cm) is: 1

(a) 102 **(b)** 69 **(c)** 120 **(d)** 123

9. **Assertion (A):** A triangle with sides 6 cm, 8 cm and 10 cm is right-angled. 1

Reason (R): If the square of the longest side equals the sum of the squares of the other two sides, the triangle is right-angled.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $(5, 12, 13)$ is a Pythagorean triplet. 1
Reason (R): $(3, 4, 5)$ is a Pythagorean triplet.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The legs of a right triangle are in the ratio $3 : 4$ and the hypotenuse is 35 cm. Find the legs. 2

12. Find a Pythagorean triplet in which one member is 19. 2

13. A kite string 50 m long is stretched straight and the kite is 48 m above the boy's hand. Find the horizontal distance of the boy from the point directly below the kite. 2

14. The legs of a right triangle are in the ratio $5 : 12$ and the hypotenuse is 65 cm. Find the legs. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 37 m long, with its foot in a street, reaches a window 35 m high on one side. Without moving its foot it is turned to reach a window 12 m high on the opposite side. Find the width of the street. 3

16. A rhombus has perimeter 292 cm and one diagonal 96 cm. Find the other diagonal and the area. 3

17. Two ships leave a port, one sailing east at 8 km/h and the other south at 15 km/h. How far apart are they after 3 hours? 3

OR

A ladder 53 m long, with its foot in a street, reaches a window 45 m high on one side. Without moving its foot it is turned to reach a window 28 m high on the opposite side. Find the width of the street. 3

18. A rhombus has perimeter 340 cm and one diagonal 26 cm. Find the other diagonal and the area. 3

OR

Two ships leave a port, one sailing east at 12 km/h and the other south at 35 km/h. How far apart are they after 2 hours? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A 13 m ladder leans against a vertical wall with its foot 5 m from the wall. 5
(i) How high up the wall does the ladder reach?
(ii) Which Pythagorean triplet does this give?
(iii) If the foot is pulled out to 12 m from the wall, how high does the ladder reach?
(iv) By how much does the top of the ladder slip down?

20. In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 5 cm, BC = 12 cm, CD = 84 cm and DA = 85 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

OR

A ladder 65 m long reaches a point 56 m up a wall. Its top slides down 4 m. Find 5
(i) the original distance of the foot from the wall,
(ii) its new distance from the wall,
(iii) how far the foot moves.

END OF QUESTION PAPER — Total: 40 Marks